

Who (Proposed Client)

FitTrack Labs is a fictional wearable health analytics company that develops fitness trackers and companion mobile apps. Their goal is to improve user engagement and daily wellness outcomes.

Why (Question/Topic being investigated)

Many FitTrack users set daily calorie-burn or activity goals, but often fail to meet them because they only assess progress late in the day, making it too late to take corrective action. This lack of timely feedback has been linked to decreased motivation and long-term user churn.

To address this issue, a goal-setting strategy can be implemented that utilizes early-day activity and health data, such as steps taken, sleep quality, heart rate metrics, and past activity trends. By analyzing this data, users can predict their end-of-day total calorie burn and determine whether they are *on track* or *at risk of missing* their goal. This enables timely, actionable guidance such as recommending a short walk before the day ends.

How (Plan of Attack)

We will use a publicly available wearable health dataset containing features such as daily steps, active minutes, calories burned, distances, sleep duration etc. After cleaning and preprocessing, we will train and compare three supervised regression models to determine which model best predicts daily calorie expenditure. We will perform model evaluation using metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) to determine which model performs best. After selecting the strongest model, we will define a simple rule-based recommendation layer that converts the predicted calorie shortfall into an estimate of how many minutes of light/moderate activity are needed to reach the daily goal.

What (Dataset, Models, Framework, Components)

- a) **Dataset:** Kaggle - FitLife: Health & Fitness Tracking Dataset

<https://www.kaggle.com/datasets/jijagallery/fitlife-health-and-fitness-tracking-dataset>

Or

Kaggle - Fitness Track Daily Activity Dataset

<https://www.kaggle.com/datasets/yaminh/fitness-track-daily-activity-dataset/data>

- b) **Models**

- i) Linear Regression
- ii) Random Forest Regressor
- iii) Gradient Boosting Regressor

- c) **Framework**

Preprocessing → Feature Engineering → Train/Validation/Test split → Train baseline linear model → Train & tune Random Forest and Gradient Boosting models → Compare model performance → Convert calorie shortfall to suggested activity recommendation → Visualize key findings (feature importance, prediction error distributions)

- d) **Components**

Preprocessing: Missing value imputation, outlier filtering, scaling numeric features (for linear model only)

Feature Engineering: Rolling averages (1, 3, 7-day), morning-only feature subset, day-type indicators (weekday/weekend)

Hyperparameter Tuning: Grid search or randomized search for tree-based models

Model Validation: GroupKFold or leave-one-user-out to avoid leakage

Visualization: Feature importance plots, predicted vs. true calorie burn plots, model comparison charts.